

**MODEL 010C
WIND SPEED SENSOR
OPERATION MANUAL**



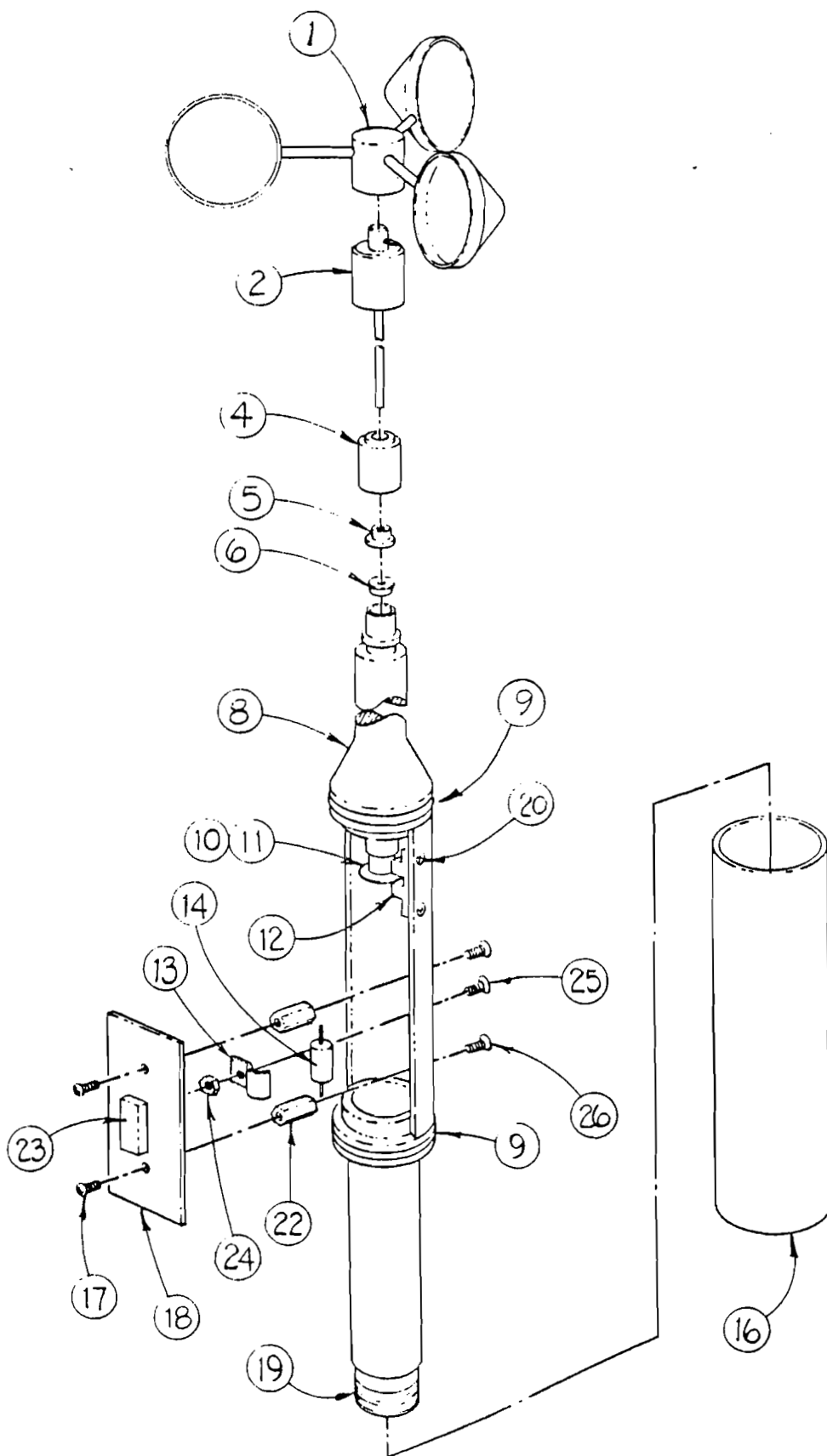
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Table 4.2
REPLACEABLE 010C PARTS LISTS

<u>Ref No.</u>	<u>Description</u>	<u>Part No.</u>
1.	Cup Assembly	Lexan-2672 Alum. - 2672-1
2.	Hub/Shaft Assy	2658
4.	Shield	1009
5.	Slinger	1010
6.	Bearing	1055
9.	O-Ring	720120
10.	Chopper Wheel Assembly	2202
11.	Set Screw	601250
12.	Photodetector Assembly	1074
13.	Heater Clamp	480100
14.	Heater	805080
16.	Sensor Cover	2675
17.	PCBA Mounting Screws	601240
18.	PCBA	1200
20.	Detector Assembly Mounting Screws	601270
22.	Standoff	860050
23.	Integrated Amplifier	620300
24.	Nut, Hex, Kep 4-40	600400
25.	Screw FH 82° 4-40 x 3/8	601330
26.	Screw FH 82° 4-40 x 1/4	601240



MODEL 010C WIND SPEED SENSOR

C. Install the new bearings and assemble the sensor.

1. Install new upper and lower bearings in the column (8). Bearings should slide easily into bearing bores.
2. Install a slinger and shield (4,5) on the column assembly. Use new parts if old ones are damaged or corroded.
3. Insert the rotating hub shaft (2) into the column assembly (8), through the shield (4), slinger, and upper bearing, until it starts to protrude through the lower bearing.
4. Support the light-chopper assembly (10) with one hand and slowly push the rotating hub shaft into it until the shaft almost bottoms.
5. Tighten both special set screws on the light-chopper assembly, do not overtighten as the set screw tips will damage the shaft.
6. Rotate the sensor hub assembly (2) to ensure that it turns freely and that an endplay of about .005" exists.
7. Hold sensor vertically and make sure that the light-chopper assembly(10) is not contacting the detector assembly (12).
8. Apply small amount of silicone lubricant (Dow-Corning DC-33 or equivalent) to the sensor O-rings (9); slide the cover (16) up over the sensor and wipe off any excess lubricant.

D. Replace cup assembly and re-install (refer to Section 2.0)

- H. Rotate the sensor hub assembly to ensure that it turns freely and that the sensor bearings are not damaged. Make sure the light-chopper assembly is not contacting the light source and detector.
- I. Apply a small amount of silicone lubricant (Dow-Corning DC-33 or equivalent) to the sensor O-ring seals; slide the cover up over the sensor and wipe off any excess lubricant.
- J. A moisture vent is located on the base of the sensor, make sure that this vent is clear.
- K. Re-install sensor according to installation procedure (Section 2.0); verify proper operation using procedures in Section 3.0.

4.3 010C Wind Speed Sensor Maintenance. (Refer to 010C Sensor Assembly Drawing.)

The following procedures require a relatively clean, dry work area, a source of 12VDC power at approximately 20 mA, and an oscilloscope (DC to 10 KHz minimum range required).

4.4 Sensor Bearing Replacement. (Refer to 010C Sensor Assembly Drawing).

- A. Remove sensor from tower and remove cup assembly (1). Refer to Section 4.2.
- B. Disassemble sensor and remove old bearings (6).
 - 1. Slide the sensor cover (16) down to expose the light-chopper disc assembly (10), detector assembly (12) and circuit board (18).
 - 2. Loosen both special set screws on the shaft of the light chopper assembly (11).
 - 3. Support the light-chopper assembly (10) with one hand and slowly pull the rotating hub/shaft assembly (2) out of the assembly (8).
 - 4. Remove the shield (4) and slinger (5) from the column (8).
 - 5. Remove the light-chopper assembly (10) from the sensor housing, being careful not to damage the slots located between the light-chopper holder and the lower bearing.
 - 6. Insert the lower end of the rotating hub/shaft assembly into the upper bearing, cock it slightly to one side and push out the lower bearing.
 - 7. Insert a right-angle type of tool, such as an Allen wrench, into the upper bearing; cock it slightly to one side and remove the bearing.
 - 8. Clean dirt from bearing bores, using a cotton swab and alcohol.

4.7 Integrated Amplifier Replacement

- A. Remove sensor from tower and remove cup assembly (refer to Section 4.2)
- B. Slide the sensor cover (16) down to expose the light-chopper disc assembly (10), detector assembly (12) and circuit board (18).
- C. Carefully pry amplifier (23) from its socket on the circuit board noting Pin 1 orientation.
- D. Install replacement integrated amplifier by reversing above procedures.

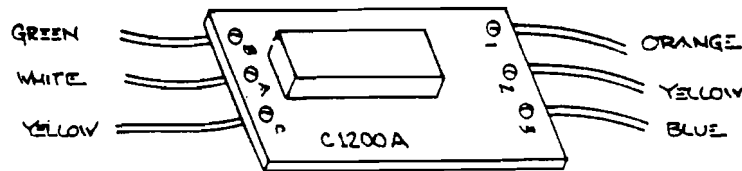
4.8 010C Wind Speed Sensor Repair and Recalibration Service.

The factory provides fast, economical service for the user. This repair and calibration service includes disassembly and detailed inspection of all moving mechanical parts and electronic components.

Service includes replacement of bearings, regardless of apparent condition, and functional test of sensor. Replacement of the following items is also included: O-rings, shield and slinger, shaft, set screws. Other components will be replaced as required. Only charges for additional materials will be added to the basic service charge.

4.5 C1200 Circuit Board Assembly Replacement. (Refer to 010C Assembly Schematic.)

- A. Remove sensor from tower and remove cup assembly (refer to Section 4.2).
- B. Slide the sensor cover (16) down to expose the light-chopper disc assembly (10), detector assembly (12) and circuit board (18).
- C. Remove two screws (17) holding circuit board assembly (18) and lift circuit board away from sensor housing.
- D. Note color of wires and then unsolder three wires to detector assembly from circuit board and three wires from connector (19).
- E. Install new circuit board assembly by reversing above procedure.



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4.6 Detector Assembly Replacement. (Refer to 010C Assembly Schematic.)

- A. Remove sensor from tower and remove cup assembly. Refer to Section 4.2.
- B. Slide the sensor cover (16) down to expose the light-chopper disc assembly (10), detector assembly (12) and circuit board (18).
- C. Remove two screws (17) holding circuit board assembly (18) and lift circuit board away from sensor housing.
- D. Note color of wires and then unsolder three wires to detector assembly from circuit board (18).
- E. Remove two screws (20) holding detector assembly (12) and remove assembly.
- F. Install new detector assembly by reversing above procedures.

MODEL 010C WIND SPEED SENSOR

1.0 GENERAL INFORMATION

- 1.1 The 010C Wind Speed Sensor uses a durable, three-cup anemometer assembly and solid-optical link with a 40-slot chopper disk to produce a pulsed output whose frequency is proportional to wind speed. An internal heater reduces moisture for extended bearing life. This sensor is usually used in conjunction with the 191 Crossarm Assembly. The sensor may be used with a translator module, or used directly with a variety of data loggers.
- 1.2 The Sensor Cable has a quick-connect connector with vinyl jacketed, shielded cable. Cable length is given in -XX feet on each cable part number. A 1953-XX cable is used with translators having terminal strip connections.

Table 1-1
Model 010C Wind Speed Sensor Specifications

Performance Characteristics

Maximum Operating Range	0-125 mph or 0-60 meters/sec.
Starting Speed	0.6 mph or .27 meters/sec
Calibrated Range	0-100 mph or 0-50 meters/sec
Accuracy	±1% or 0.15 mph
Temperature Range	-50°C to +85°C
Response	Distance constant less than 5 feet* of flow

*The distance traveled by the air after a sharp-edged gust has occurred for the anemometer rate to reach 63% of the new speed.

Electrical Characteristics

Power Requirements	12VDC at 10 mA
Output Signal	11-volt pulse
Output Impedance	100 ohms maximum
Heater Power Requirement	12VDC at 350 mA
Standard Cable Length	300' maximum (consult factory if longer cable is to be used for special requirements)

Physical Characteristics

Weight	1.1 pounds
Finish	Anodized aluminum
Mounting	Use with 191 Mounting Arm
Cabling	1953-XX Cable (XX is cable length in feet)

Optional Accessories

- a. External heater and power supply for extreme low temperature operation.
- b. Ice Skirt for extreme icing environments.
- c. Aluminum cup assembly.

2.0 INSTALLATION

- 2.1 Mount the cup assembly and secure with the allen head set screw, check to see that the cup assembly rotates freely.
- 2.2 Install the sensor in the end of the Model 191 Mounting Arm (the end without the bushing).
- 2.3 Tighten the locking set screw. Do not over-tighten. Apply a small amount of silicone grease to set screws to prevent freezing in corrosive environments.
- 2.4 Connect the cable assembly to the keyed sensor receptacle and tape it to the mounting arm.
- 2.5 Wiring. The cable assembly contains five wires. Typical wiring hookup is shown in Figure 2-1.
- 2.6 Lightning Protection. Weather sensors are sensitive to direct or nearby lightning strikes. A well-grounded metal rod or frame should be placed above the sensor installation. In addition, the shield on the signal cable leading to the translator must be connected to a good earth ground at the translator end and the cable route should not be vulnerable to lightning.

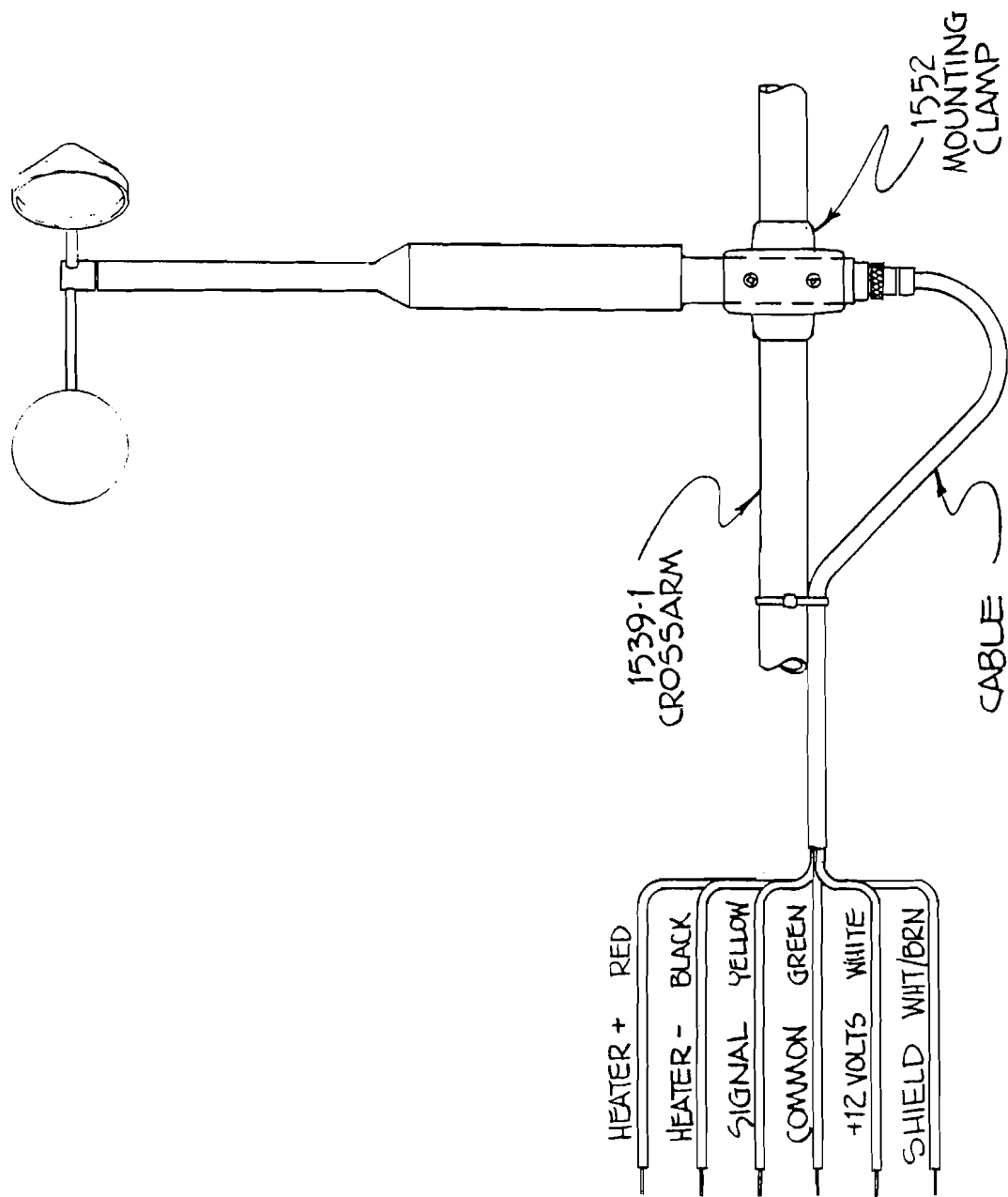


FIG. 2-1
TYPICAL OIOC INSTALLATION

3.0 OPERATIONAL CHECK-OUT AND CALIBRATION

- 3.1 Spinning the anemometer cup assembly will produce a series of pulses (40 pulses per revolution). To verify sensor output, monitor this signal with either the translator module, data logger or an oscilloscope. (Refer to Frequency Vs. Wind Speed, Table 3-1.) Spinning the hub of the wind speed transmitter without the cup assembly mounted and allowing it to coast to a stop will give a good indication of threshold performance; a jerky or sudden stop indicates damage to bearings, bent drive shaft or obstruction in the light chopper.
- 3.2 Inspect the cup assembly for loose cup arms or other damage. The cup assembly cannot change calibration unless a mechanical part has come loose or has been broken. If a cup arm is loose or broken the calibration of the sensor may be affected.
- 3.3 If the sensor heater is used, check internal heater operation by sliding sensor cover down and touching the housing behind the printed circuit board. The housing should feel warmer than the adjoining metal parts. The sensor has a built-in heater that is designed to provide a raise in the internal temperature, providing a small positive pressure. This heater requires an external 12V (@500ma) power supply.

4.0 MAINTENANCE AND TROUBLESHOOTING

4.1 General Maintenance Schedule*

6-12 Month Intervals:

- A. Inspect sensor for proper operation per Section 4.2.
- B. Replace wind speed sensor bearings in extremely adverse environments per Section 4.5.

12-24 Month Intervals:

- A. Recommended complete factory overhaul of sensor.

*Schedule is based on average to adverse environments.

FREQUENCY VS WIND SPEED FOR 010 SENSOR
TABLE 3-1

THRESHOLD = 0.27 MPS OR 0.6 MPH

MPH = 0.44704 MPS

MPH	F (Hz)	MPS	RPS	MPS	F (Hz)	MPH	RPS
10.0	105.0732	4.4704	2.62683	2.5	55.78355	5.592340	1.394588
20.0	216.8532	8.9408	5.42133	5.0	118.3211	11.18468	2.958029
30.0	328.6332	13.4112	8.21583	7.5	180.8587	16.77702	4.521469
40.0	440.4132	17.8816	11.01033	10.0	243.3963	22.36936	6.084909
50.0	552.1932	22.352	13.80483	12.5	305.9340	27.96170	7.648350
60.0	663.9732	26.8224	16.59933	15.0	368.4716	33.55404	9.211790
70.0	775.7532	31.2928	19.39383	17.5	431.0092	39.14638	10.77523
80.0	887.5332	35.7632	22.18833	20.0	493.5468	44.73872	12.33867
90.0	999.3132	40.2336	24.98283	22.5	556.0844	50.33106	13.90211
100.0	1111.093	44.704	27.77733	25.0	618.6220	55.92340	15.46555
110.0	1222.873	49.1744	30.57183	27.5	681.1597	61.51574	17.02899
120.0	1334.653	53.6448	33.36633	30.0	743.6973	67.10808	18.59243
130.0	1446.433	58.1152	36.16083	32.5	806.2349	72.70042	20.15587

RPM	MPS	MPH	F (Hz)
100	2.935062	6.564096	66.66666
200	5.600125	12.52819	133.3333
300	8.265188	18.49228	200
400	10.93025	24.45638	266.6666
500	13.59531	30.42048	333.3333
600	16.26037	36.38457	400
700	18.92543	42.34867	466.6666
800	21.59050	48.31276	533.3333
900	24.25556	54.27686	600
1000	26.92062	60.24096	666.6666
1100	29.58568	66.20505	733.3333
1200	32.25075	72.16915	800
1300	34.91581	78.13324	866.6666
1400	37.58087	84.09734	933.3333
1500	40.24594	90.06144	1000
1600	42.91100	96.02553	1066.666
1700	45.57606	101.9896	1133.333
1800	48.24112	107.9537	1200

For MPS:

$$RPS = 0.62537615 * (V \text{ MPS} - 0.27)$$

$$V \text{ MPS} = (Hz * .03997594) + 0.27$$

$$Hz = RPS * 40 = \frac{V - 0.27}{0.039977}$$

$$RPM = 37.552569 * (V \text{ MPS} - 0.27)$$

$$V \text{ MPS} = (RPM / 37.552569) + 0.27$$

For MPH:

$$RPS = 0.27945 * (V \text{ MPH} - 0.6)$$

$$V \text{ MPH} = (Hz * .08946144) + 0.6$$

$$Hz = RPS * 40$$

$$RPM = 16.767 * (V \text{ MPH} - 0.6)$$

$$V \text{ MPH} = (RPM / 16.767) + 0.6$$

Table 4-1
010C Wind Speed Sensor Troubleshooting

<u>Symptom</u>	<u>Probable Cause</u>	<u>Solution</u>	<u>Refer to</u>
No wind speed output	Loss of supply voltage	Check translator +12 supply & connecting cables	FIG 2-1
	Faulty integrated amplifier	Replace IC amplifier(plug-in)	SECT 4.7
	Faulty diodes, D1, D2	Replace circuit board	SECT 4.5
	Faulty detector	Replace detector	SECT 4.6
No wind speed output below 2-5 mph	Bad bearings	Replace bearings	SECT 4.4
	Faulty detector	Replace detector	SECT 4.6
Wind speed signal drops out as speed increases	Faulty detector	Replace detector	SECT 4.6

4.2 010C Wind Speed Sensor: 6-12 Month Periodic Service

- A. At the crossarm assembly, disconnect the quick disconnect plug from the sensor (leave the cable secured to the crossarm) and remove the sensor from the crossarm assembly.
- B. Loosen the set screw holding the cup assembly. Support the rotating hub of the sensor with one hand and pull the anemometer cup assembly free.
- C. Visually inspect the anemometer cups for cracks and breaks, and make sure that each arm is securely attached to the cup assembly hub.
- D. Slide the sensor cover down to expose the light-chopper disc assembly, light source and detector and circuit board.
- E. Inspect the interior of the sensor for any signs of corrosion and dust buildup.
- F. Inspect the light-chopper for cracks, and make sure that all slots are free of corrosion.
- G. Inspect the signal conditioning module for cracks and corrosion around soldered connections.

- H. Rotate the sensor hub assembly to ensure that it turns freely and that the sensor bearings are not damaged. Make sure the light-chopper assembly is not contacting the light source and detector.
- I. Apply a small amount of silicone lubricant (Dow-Corning DC-33 or equivalent) to the sensor O-ring seals; slide the cover up over the sensor and wipe off any excess lubricant.
- J. A moisture vent is located on the base of the sensor, make sure that this vent is clear.
- K. Re-install sensor according to installation procedure (Section 2.0); verify proper operation using procedures in Section 3.0.

4.3 010C Wind Speed Sensor Maintenance. (Refer to 010C Sensor Assembly Drawing.)

The following procedures require a relatively clean, dry work area, a source of 12VDC power at approximately 20 mA, and an oscilloscope (DC to 10 KHz minimum range required).

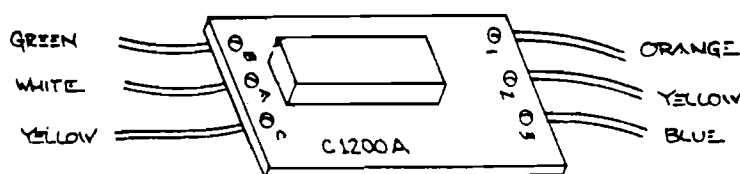
4.4 Sensor Bearing Replacement. (Refer to 010C Sensor Assembly Drawing).

- A. Remove sensor from tower and remove cup assembly (1). Refer to Section 4.2.
- B. Disassemble sensor and remove old bearings (6).
 - 1. Slide the sensor cover (16) down to expose the light-chopper disc assembly (10), detector assembly (12) and circuit board (18).
 - 2. Loosen both special set screws on the shaft of the light chopper assembly (11).
 - 3. Support the light-chopper assembly (10) with one hand and slowly pull the rotating hub/shaft assembly (2) out of the assembly (8).
 - 4. Remove the shield (4) and slinger (5) from the column (8).
 - 5. Remove the light-chopper assembly (10) from the sensor housing, being careful not to damage the slots located between the light-chopper holder and the lower bearing.
 - 6. Insert the lower end of the rotating hub/shaft assembly into the upper bearing, cock it slightly to one side and push out the lower bearing.
 - 7. Insert a right-angle type of tool, such as an Allen wrench, into the upper bearing; cock it slightly to one side and remove the bearing.
 - 8. Clean dirt from bearing bores, using a cotton swab and alcohol.

- C. Install the new bearings and assemble the sensor.
1. Install new upper and lower bearings in the column (8). Bearings should slide easily into bearing bores.
 2. Install a slinger and shield (4,5) on the column assembly. Use new parts if old ones are damaged or corroded.
 3. Insert the rotating hub shaft (2) into the column assembly (8), through the shield (4), slinger, and upper bearing, until it starts to protrude through the lower bearing.
 4. Support the light-chopper assembly (10) with one hand and slowly push the rotating hub shaft into it until the shaft almost bottoms.
 5. Tighten both special set screws on the light-chopper assembly, do not overtighten as the set screw tips will damage the shaft.
 6. Rotate the sensor hub assembly (2) to ensure that it turns freely and that an endplay of about .005" exists.
 7. Hold sensor vertically and make sure that the light-chopper assembly(10) is not contacting the detector assembly (12).
 8. Apply small amount of silicone lubricant (Dow-Corning DC-33 or equivalent) to the sensor O-rings (9); slide the cover (16) up over the sensor and wipe off any excess lubricant.
- D. Replace cup assembly and re-install (refer to Section 2.0)

4.5 C1200 Circuit Board Assembly Replacement. (Refer to 010C Assembly Schematic.)

- A. Remove sensor from tower and remove cup assembly (refer to Section 4.2).
- B. Slide the sensor cover (16) down to expose the light-chopper disc assembly (10), detector assembly (12) and circuit board (18).
- C. Remove two screws (17) holding circuit board assembly (18) and lift circuit board away from sensor housing.
- D. Note color of wires and then unsolder three wires to detector assembly from circuit board and three wires from connector (19).
- E. Install new circuit board assembly by reversing above procedure.



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4.6 Detector Assembly Replacement. (Refer to 010C Assembly Schematic.)

- A. Remove sensor from tower and remove cup assembly. Refer to Section 4.2.
- B. Slide the sensor cover (16) down to expose the light-chopper disc assembly (10), detector assembly (12) and circuit board (18).
- C. Remove two screws (17) holding circuit board assembly (18) and lift circuit board away from sensor housing.
- D. Note color of wires and then unsolder three wires to detector assembly from circuit board (18).
- E. Remove two screws (20) holding detector assembly (12) and remove assembly.
- F. Install new detector assembly by reversing above procedures.

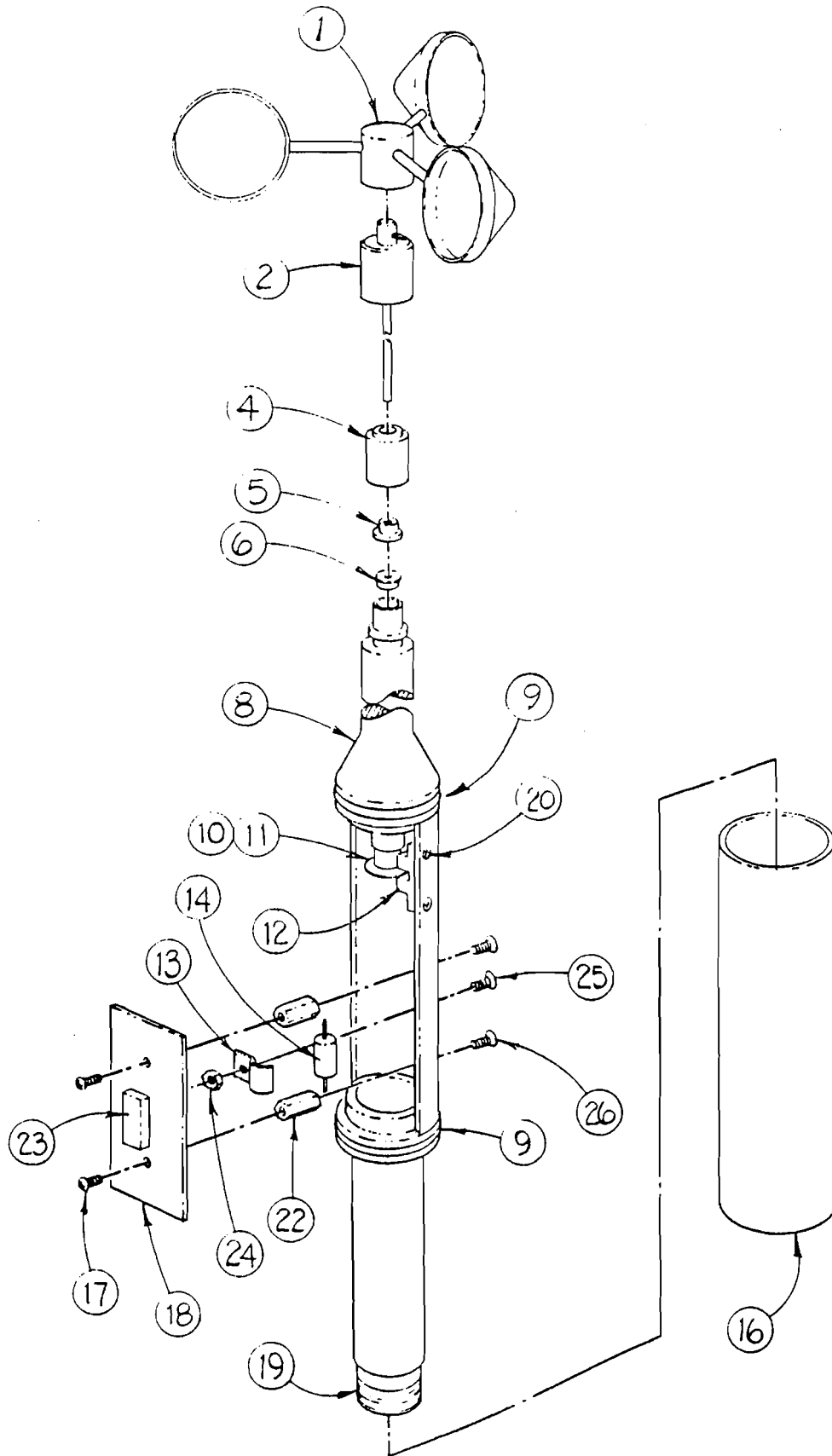
4.7 Integrated Amplifier Replacement

- A. Remove sensor from tower and remove cup assembly (refer to Section 4.2)
- B. Slide the sensor cover (16) down to expose the light-chopper disc assembly (10), detector assembly (12) and circuit board (18).
- C. Carefully pry amplifier (23) from its socket on the circuit board noting Pin 1 orientation.
- D. Install replacement integrated amplifier by reversing above procedures.

4.8 010C Wind Speed Sensor Repair and Recalibration Service.

The factory provides fast, economical service for the user. This repair and calibration service includes disassembly and detailed inspection of all moving mechanical parts and electronic components.

Service includes replacement of bearings, regardless of apparent condition, and functional test of sensor. Replacement of the following items is also included: O-rings, shield and slinger, shaft, set screws. Other components will be replaced as required. Only charges for additional materials will be added to the basic service charge.

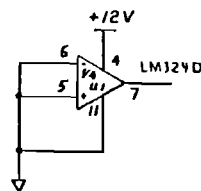
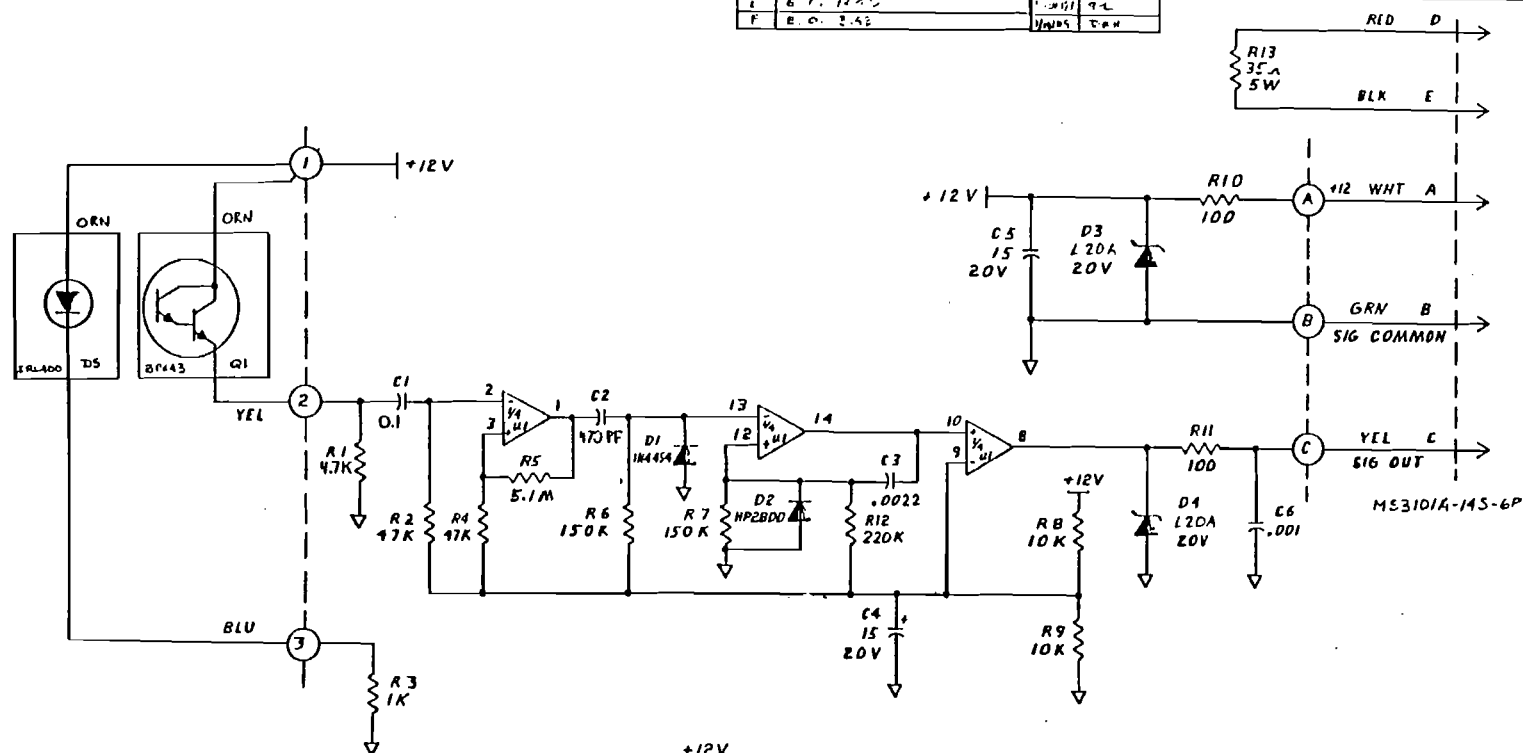


MODEL 010C WIND SPEED SENSOR

Table 4.2
REPLACEABLE 010C PARTS LISTS

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18.	PCBA	1200
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23.	Integrated Amplifier	620300
24.	Nut, Hex, Kep 4-40	600400
25.	Screw FH 82° 4-40 x 3/8	601330
26.	Screw FH 82° 4-40 x 1/4	601240

REVISIONS			DATE	APPROVAL
B	CHANGE C2 WAS 100PF	13 39	12/7/75	
I	A.I. 200K TO 220K	12 47	12/7/75	
E	G.F. 12K	12 47	12/7/75	
F	B.C. 2.42	12 47	12/7/75	



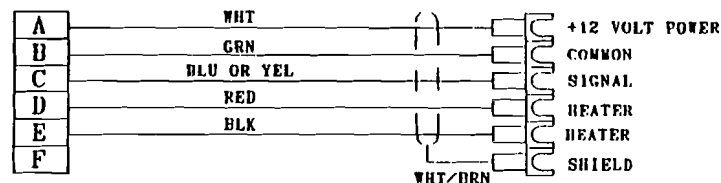
NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL RESISTORS ARE IN OHMS. $\pm 5\%$, 1/4W, CARBON
2. ALL 1% RESISTORS ARE RISED
3. ALL CAPACITORS ARE IN MFD.
4. REFERENCE DESIGNATORS LAST USED

R 13 0 K P TP
C 6 0 2 S F
D 4 L T T

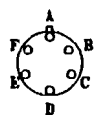
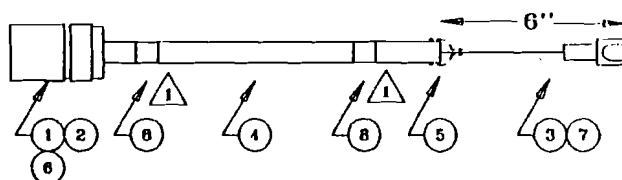
MATERIAL			FINISH		HEAT TREAT	
DIMENSIONAL TOLERANCES UNLESS OTHERWISE NOTED			TITLE		WEIGHT	
FRAC	DEC	ANG.	SCHEMATIC, WIND SPEED HEAD			
1	2	3	DRAWN F. COLLUM DATE 10-2-75		SIZE	
1	2	3	CHECKED H. KELLOGG DATE 12-15-75		DRAWING NUMBER	
1	2	3	APPROVED DATE		ISSUE	
1	2	3			C 1202 F	

REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
D	REDRAWN PER E.O. 1131	9/11/91	DH
E	WIRE COLORS	10/17/91	DH



SENSOR END

TRANSLATOR END



SOLDER CUP VIEW

△ IDENTIFY CABLE 16" FROM EACH END.
DASH NUMBER = LENGTH IN FEET.

ITEM	PART NO.	DESCRIPTION	QTY
1	500351	CONNECTOR, 6 PIN, FEMALE	1
2	480500	CLAMP	1
3	600193	LUG, SPADE, #6	6
4	400014	CABLE, 5 COND., SHIELDED	A/R
5	060050	SLEEVING, 1/4", SHRINK	A/R
6	060075	SLEEVING, 1/8", SHRINK	A/R
7	900510	WIRE, 22 AWG, WHT/DRN	6"
8	060060	SLEEVING, 1/4", CLEAR SHRINK	A/R
9			
10			

MET ONE INSTRUMENTS			
ASSY. CABLE, 010			
WIND SPEED			
SIZE	FSCH NO.	DWG NO.	REV
		1953	E
SCALE		SHEET 1 OF 1	